

**Proposal to Replace Incandescent Exit Signs with LED Self-Illuminating Exit Signs
at Harborview Logistics, Inc.**

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Introduction and Purpose

Harborview Logistics, Inc. operates three regional distribution warehouses in Connecticut, each running around the clock to support same-day and next-day shipping commitments. Every one of these facilities still relies on incandescent exit signs installed when the buildings were constructed. This proposal recommends that Harborview replace all incandescent exit signage across its three facilities with self-illuminating light-emitting diode (LED) exit signs. The purpose of this proposal is to demonstrate the operational, financial, and safety case for the conversion and to request approval and budget authorization from the Director of Operations to begin the project in the next fiscal quarter.

Problem

Incandescent exit signs are inexpensive to purchase but expensive to own. Because exit signage must remain lit continuously, every incandescent fixture in the building draws power 24 hours a day, 365 days a year, and the bulbs inside them burn out far more quickly than modern alternatives. Facilities staff at Harborview currently replace exit sign bulbs on a rotating basis roughly every four to six weeks per warehouse, a recurring task that consumes technician labor hours that could otherwise be spent on preventive maintenance elsewhere in the building.

Beyond the labor cost, the electricity draw of incandescent exit signage is substantial relative to the light output it provides. Industry data compiled by the U.S. Department of Energy's Hospital Energy Alliance program indicates that a typical incandescent exit sign consumes roughly 350 kWh of electricity per year, compared to only about 44 kWh for an equivalent LED unit, a reduction of nearly 90% in energy draw for the same continuous-duty

application (U.S. Department of Energy, n.d.). Given that Harborview's three warehouses collectively operate more than 90 exit signs, this inefficiency represents a meaningful and entirely avoidable line item in the company's annual utility and maintenance budgets.

Finally, incandescent signage carries a hidden reliability risk. Bulb failures are unpredictable, and a burned-out exit sign is both an OSHA life-safety concern and a liability exposure for the company. The current maintenance model depends on staff noticing a failed sign during routine walkthroughs rather than on a fixture that is inherently long-lived and dependable.

Proposed Solution

Harborview should replace all existing incandescent exit sign fixtures with self-illuminating LED exit signs across its three warehouse locations. Self-illuminating LED units draw only a few watts of power, generate far less waste heat, and are rated for roughly 10 to 12 years of continuous operation before the light source needs attention, essentially eliminating the recurring bulb-replacement cycle that currently consumes facilities labor (Kolin, 2022). Because LED exit signs are solid-state devices rather than filament-based bulbs, they are also far less susceptible to the sudden, unpredictable failures that make incandescent signage a safety liability.

Installation of the new signs is straightforward. In most cases, the existing electrical box and wiring can be reused, since the new fixture simply mounts in place of the old one; industry sources describe the labor involved as comparable to hanging a picture frame (Kolin, 2022). This means the conversion can be completed by Harborview's existing facilities team over a series of scheduled maintenance windows, without hiring outside electrical contractors or disrupting warehouse operations.

Costs and Benefits

The upfront cost of the conversion is modest relative to the ongoing savings it generates. Comparable retrofit projects provide a useful benchmark: a 300-sign LED exit sign retrofit at a large hospital facility cost approximately \$17,100 in total (about \$57 per sign, including installation) and produced annual savings of roughly \$14,755 in reduced electricity and maintenance costs, for a simple payback period of just over one year (U.S. Department of Energy, n.d.).

Applying comparable per-sign figures to Harborview's roughly 90 exit signs suggests an estimated total project cost in the neighborhood of \$5,100 to \$6,500 for fixtures and installation labor. Based on the energy-savings ratio reported for similar conversions, Harborview can expect annual electricity savings of several thousand dollars, plus substantial additional savings from the near-elimination of bulb replacement labor, since LED signs typically need no lamp service for 10 or more years compared to the current 4- to 6-week replacement cycle (NV Energy, n.d.). Table 1 summarizes the estimated costs and benefits of the conversion.

Item	Estimated Value
Number of exit signs (3 warehouses)	~90
Estimated cost per sign (fixture + installation)	\$57–\$70
Estimated total project cost	\$5,100–\$6,500
Estimated annual electricity savings	~\$3,500–\$4,500
Estimated annual maintenance/labor savings	~\$1,000–\$2,000
Estimated simple payback period	Under 2 years
Expected LED sign service life	10–12 years

Even using conservative estimates, the project is expected to pay for itself in well under two years, after which the reduced electricity and labor costs become ongoing savings to the facilities budget for the remaining service life of the signs, typically a decade or more (Kolin, 2022; U.S. Department of Energy, n.d.).

Feasibility and Implementation Plan

This project is feasible within Harborview's existing operational structure. No new staff, outside contractors, or building modifications are required. The recommended implementation approach is as follows: first, the facilities team will conduct a sign count and condition audit at each of the three warehouses; second, LED replacement units will be ordered in bulk to secure a volume discount; third, installation will be scheduled in phases, one warehouse at a time, during off-peak overnight hours to avoid any disruption to shipping operations; and fourth, the facilities team will retire and responsibly recycle the incandescent fixtures that are removed. The full conversion across all three sites can realistically be completed within a single fiscal quarter.

Conclusion and Recommendation

Replacing Harborview's incandescent exit signs with self-illuminating LED units is a low-risk, low-cost project with a fast payback and a decade or more of ongoing savings and reduced safety risk. The conversion requires no new headcount, minimal capital outlay, and can be completed using existing facilities staff and infrastructure. I recommend that the Director of Operations approve a budget of up to \$7,000 to begin the LED exit sign conversion project across all three Harborview warehouses in the coming quarter.

References

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